

Bucky++

Variant: 1.0

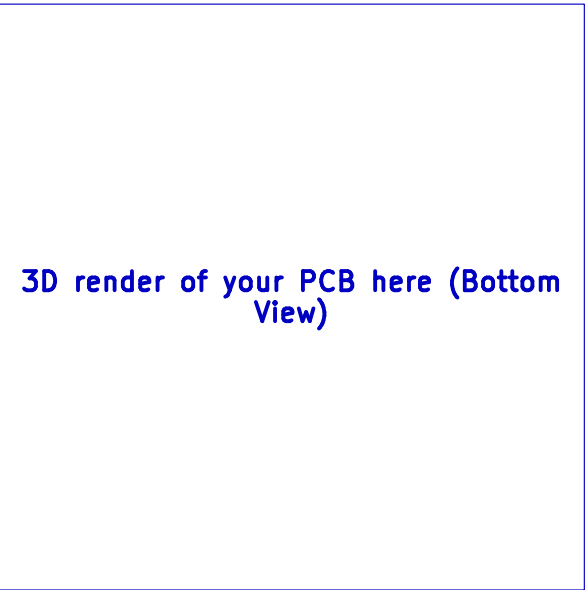
2025-12-19
Rev B

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TOP VIEW



BOTTOM VIEW



NOTES

Add a comment here

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

1.0 Release Date

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

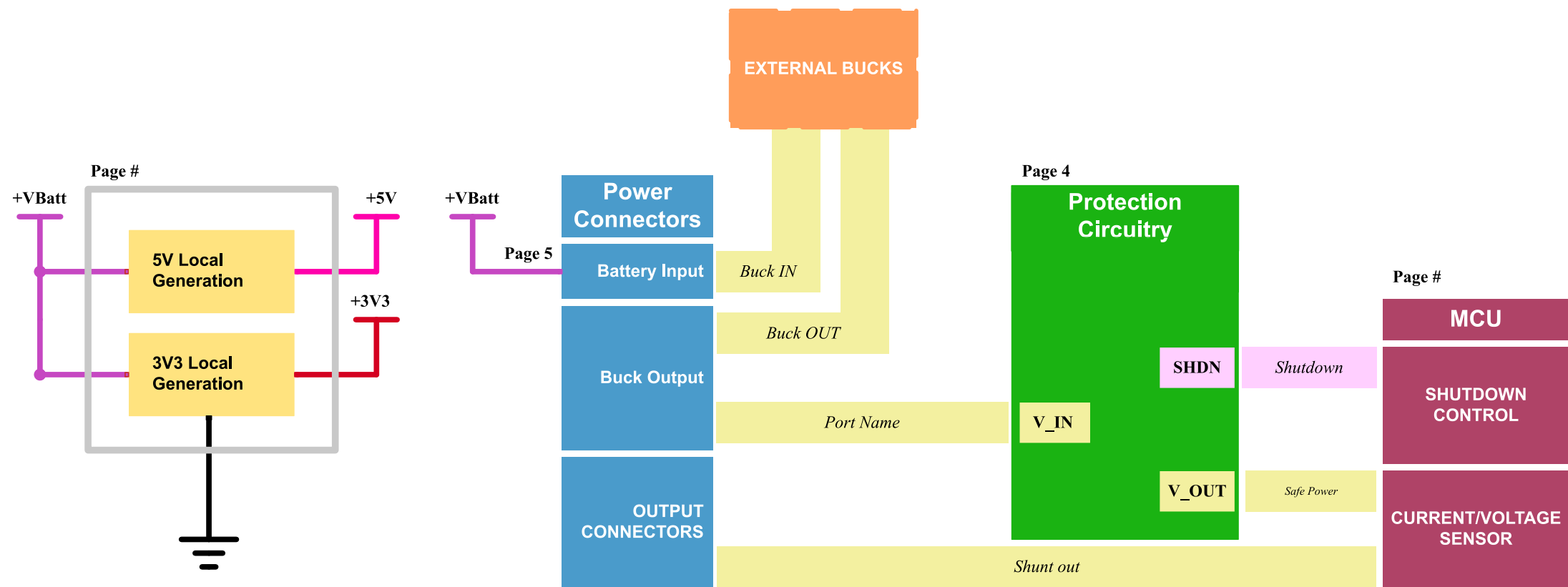
DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

	Comments:	Company: Xplore		Variant: 1.0	
		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Cover Page	File Name: Bucky++.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
	Sheet Path: /		Reviewer: Reviewer Name	Size: A3	Sheet: 1 of 12

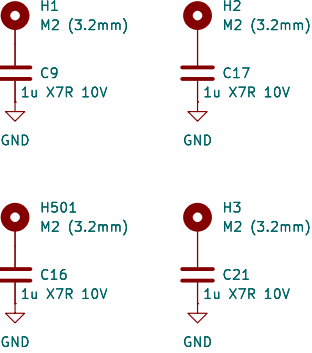
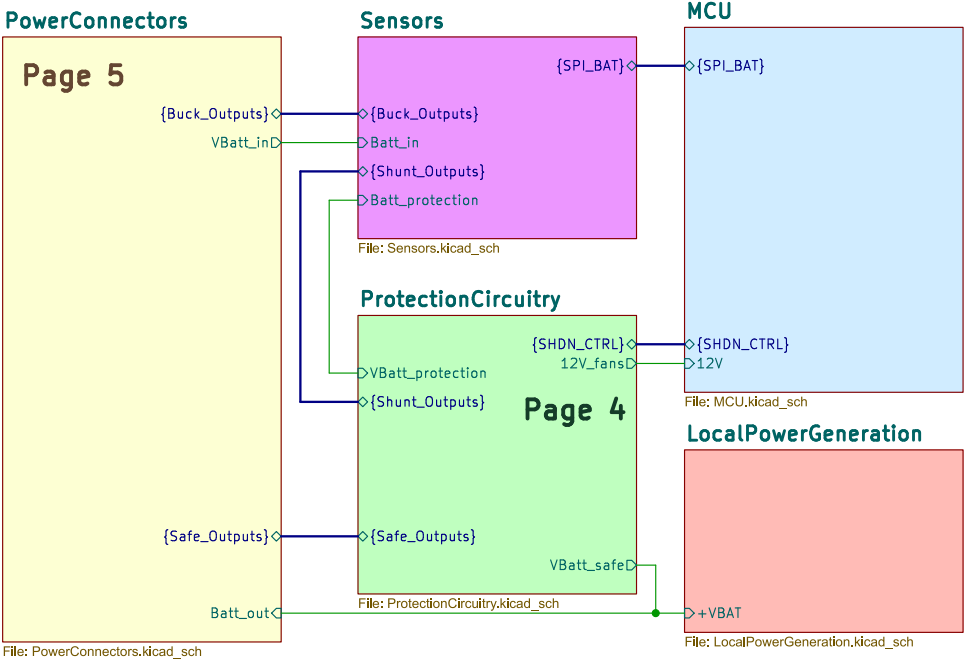
[2] Block Diagram



Target specifications:	
Input voltage:	22 - 26 V
Spec 2	??
Spec 3	??
Spec 4	??

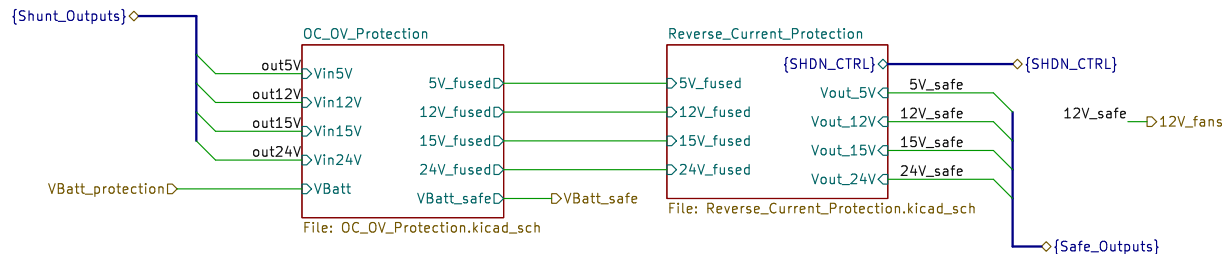
	Comments:	Company: Xplore		Variant: 1.0	
		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
Sheet Path: /Block Diagram/			Reviewer: Reviewer Name	Size: A3	Sheet: 2 of 12

[3] Project Architecture



	Comments:	Company: Xplore		Variant: 1.0	
		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
Sheet Path: /Project Architecture/			Reviewer: Reviewer Name	Size: A4	Sheet: 3 of 12

[4] Protection Circuitry



Comments:

Company:

Xplore

Variant:

1.0

Board Name:

Bucky++

Project Name:

XRC

Sheet Title:

Protection Circuitry

File Name:

ProtectionCircuitry.kicad_sch

Designer:

Alain

Date:

Last Modified Date

Revision:

B

Sheet Path:

/Project Architecture/ProtectionCircuitry/

Reviewer:

Reviewer Name

Size:

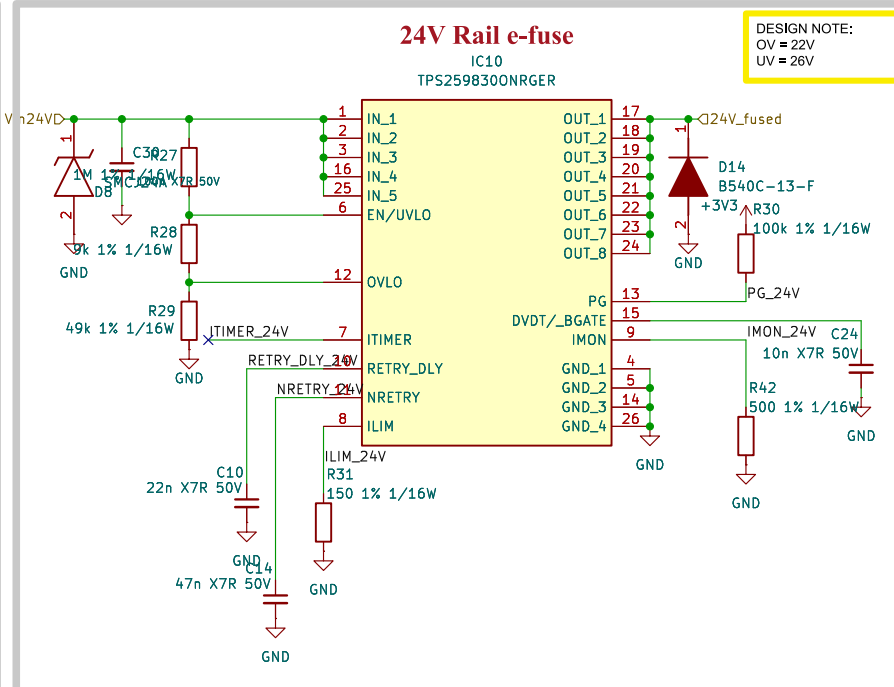
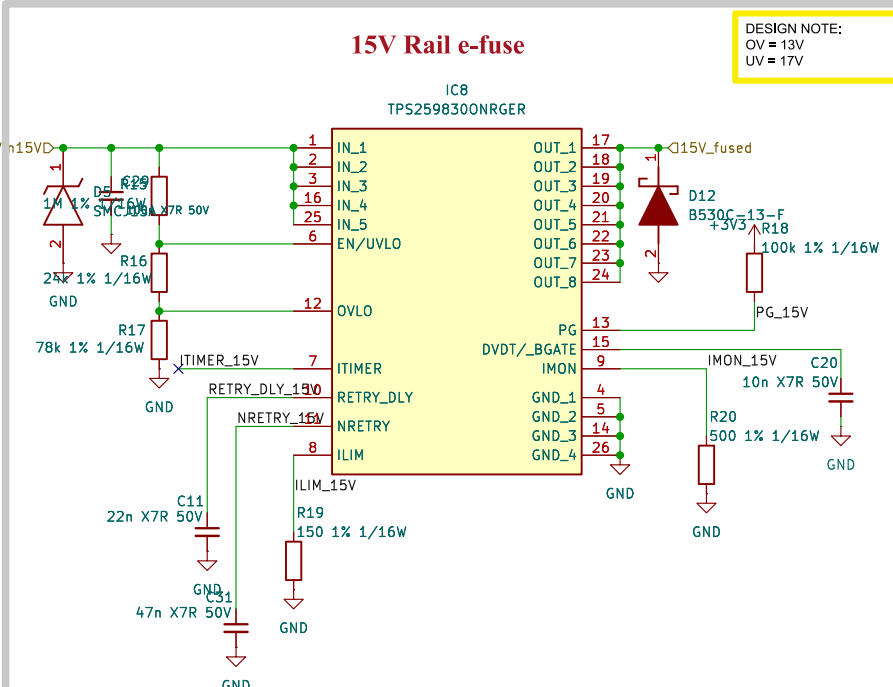
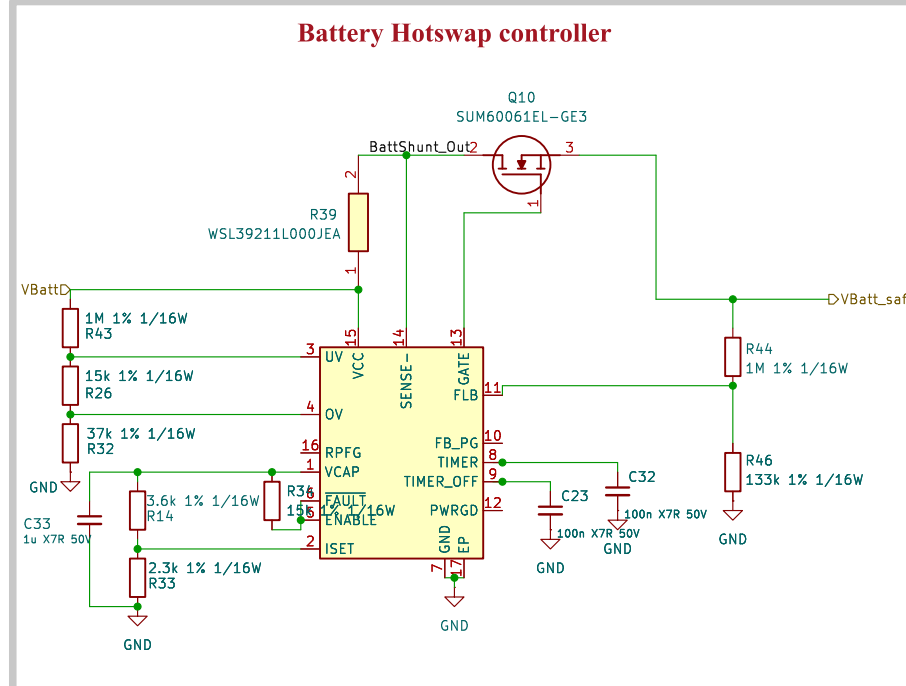
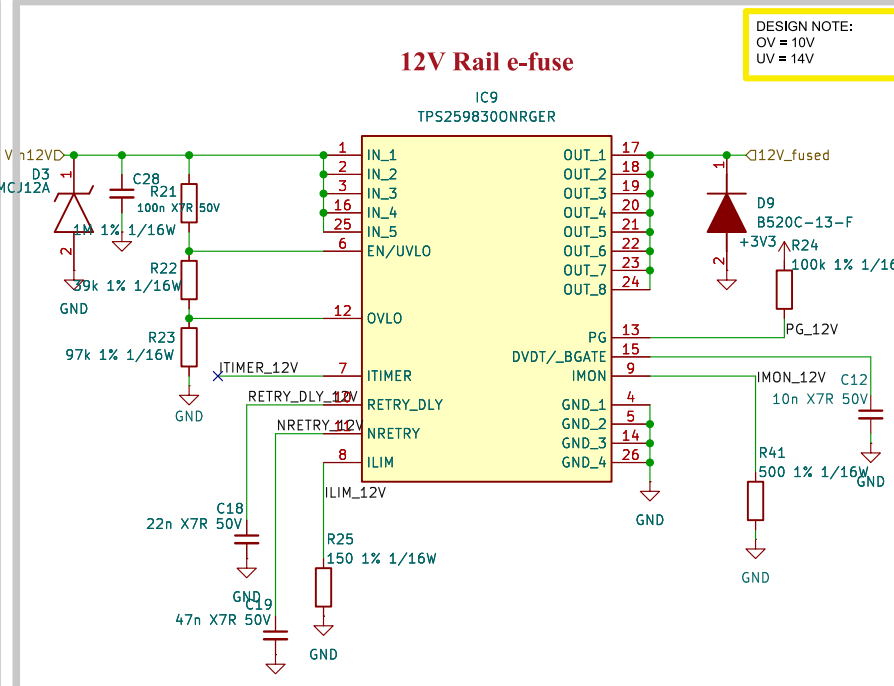
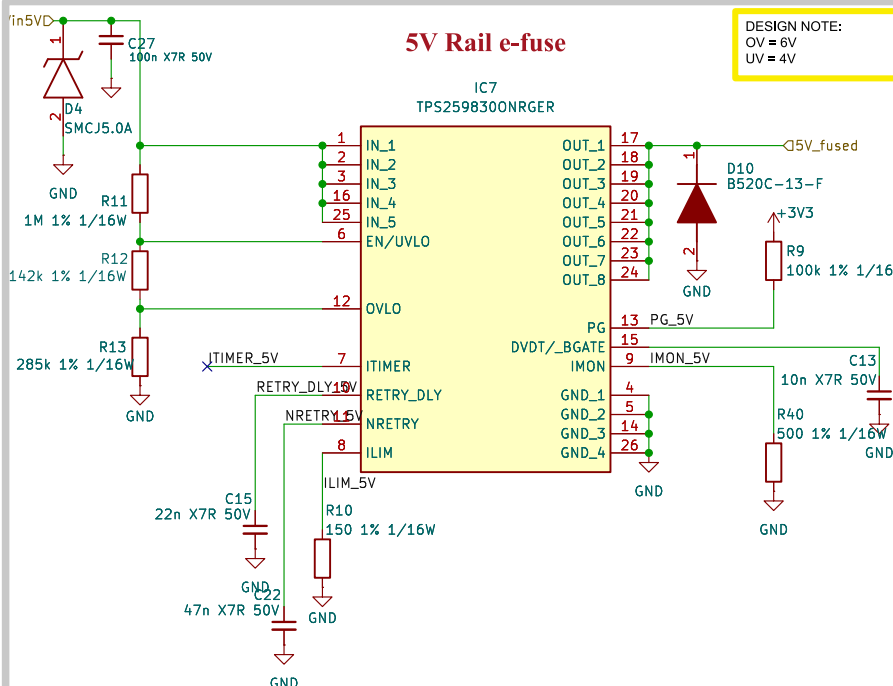
A5

Sheet:

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of 12

[11] OC_OV_Protection



LAYOUT NOTE:
The minimum trace width for 1oz copper foil is 0.02" per amp to make sure the trace stays at a reasonable temperature. 0.03" per amp or wider is recommended

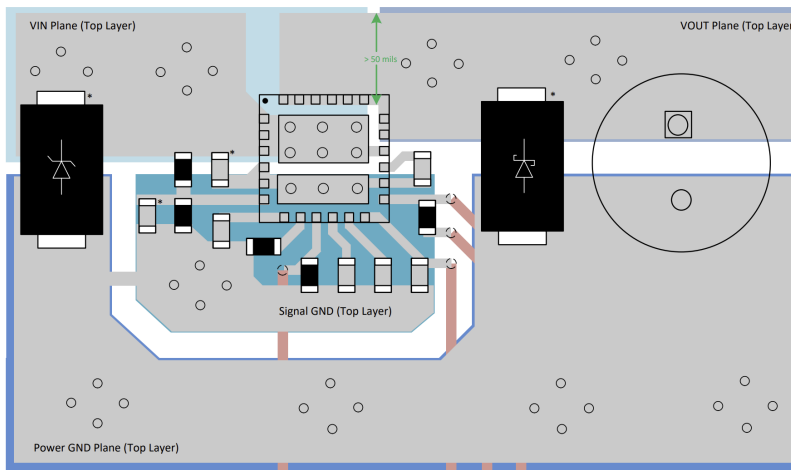


Table 7-5. NRETRY and RETRY_DLY Combination Example.

Auto Retry Delay	915 ms	416 ms	91.7 ms	9.3 ms	3 ms
RETRY_DLY Capacitor	22 nF	10 nF	2.2 nF	220 pF	68 pF
No. of Auto Retries	NRETRY Capacitor				
4	Open				
16	47 nF	22 nF	4.7 nF	1 nF	220 pF
64	0.22 μ F	0.1 μ F	22 nF	2.2 nF	1 nF
256	1 μ F	0.47 μ F	0.1 μ F	10 nF	4.7 nF
1024	3.3 μ F	1.5 μ F	0.47 μ F	33 nF	10 nF
Infinite	Short to GND				

DESIGN NOTE:
The number of time the e-fuse can retry is set at 16 and the delay between retries is maximised by using the capacitor values in the table provided by TI.

IMON pin is not used as we already have sensing, it is wired as recommended in the datasheet.

The duration of overcurrent can be set using the capacitor connected to ITimer. We want the shortest possible duration so we leave the pin

$$V_{IN(UV)} = \frac{V_{UVLO(R)} \times (R1 + R2 + R3)}{R2 + R3}$$

$$V_{IN(OV)} = \frac{V_{OV(R)} \times (R1 + R2 + R3)}{R3}$$

$$R_{\text{ILIM}}(\Omega) = \frac{1460}{I_{\text{ILIM}}(A) - 0.1}$$

$$t_{\text{TIMER}} (\text{ms}) = \frac{C_{\text{TIMER}} (\text{nF}) \times \Delta V_{\text{TIMER}} (\text{V})}{I_{\text{TIMER}} (\mu\text{A})}$$

Comment

Company:

Variant:

.0

Board Name

Project Name

Bucky++

XRC

Sheet Title

File Name

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1. **o**

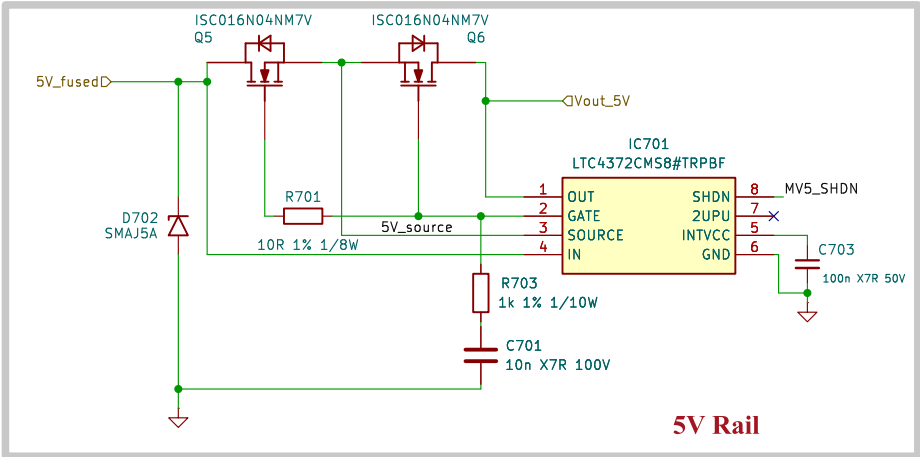
1 of 12

[12] Reverse_Current_Protection

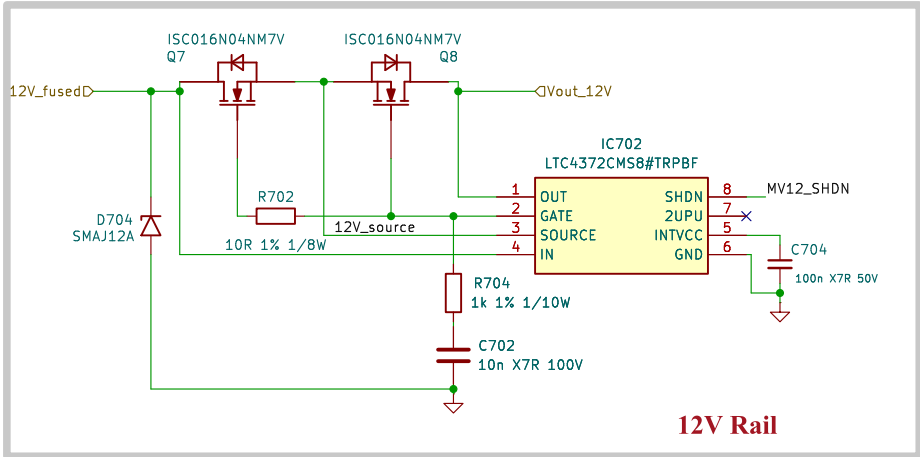
DESIGN NOTE:
TVS diodes are added to protect the chip
and MOSFETS

DESIGN NOTE:
ISC016N04NM7V was chosen as it can
hold 40V and up to 174A.

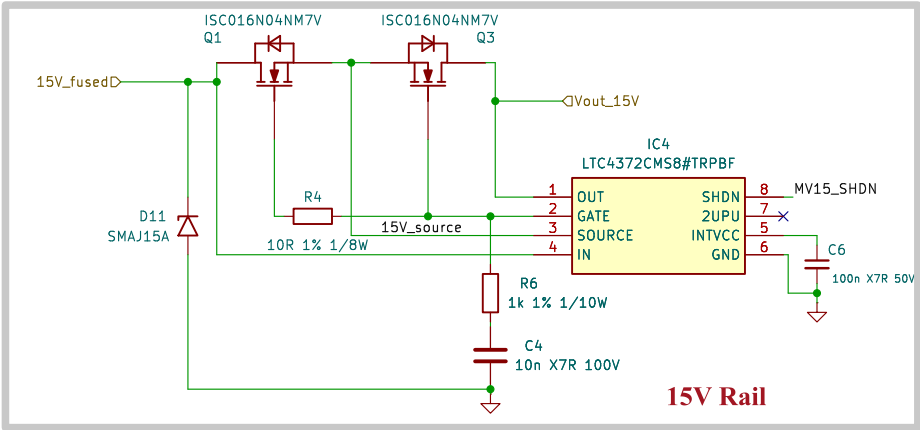
With power dissipation of over 80W, this is
more than enough even for the main input
as $P=I^2R$. This Transistor can theoretically
hold over 200A.



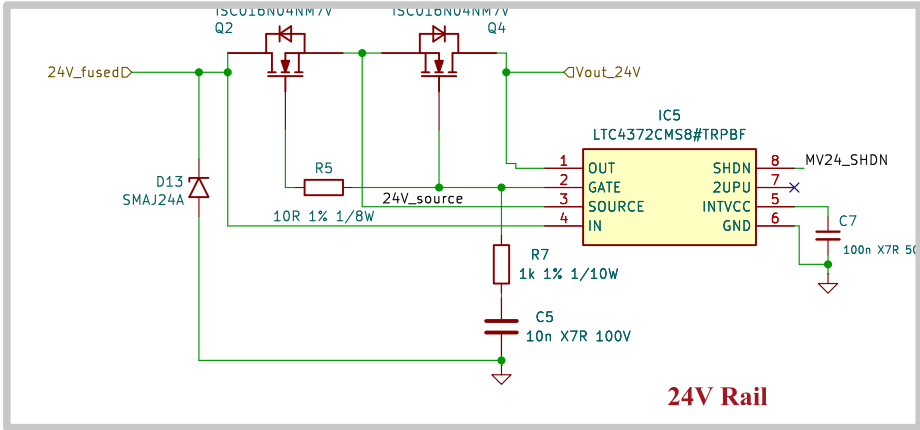
5V Rail



12V Rail



15V Rail



24V Rail

MV5_SHDN
MV12_SHDN
MV15_SHDN
MV24_SHDN

{SHDN_CTRL}

Comments:

Company:

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1.0

Board Name:

Bucky++

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XRC

Sheet Title:

Reverse_Current_Protection

File Name:

Reverse_Current_Protection.kicad_sch

Designer:

Alain

Date:

Revision:

Sheet Path:

/Project Architecture/ProtectionCircuitry/Reverse_Current_Protection/

Reviewer:

Reviewer Name

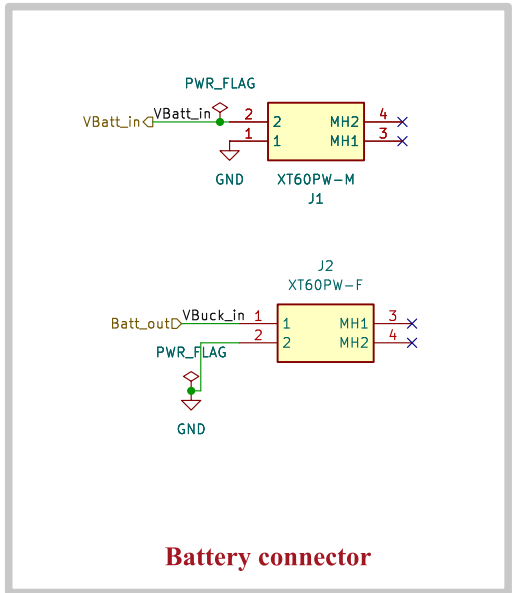
Size:

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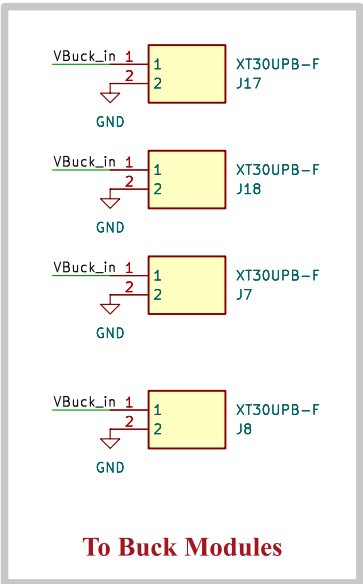
Sheet:

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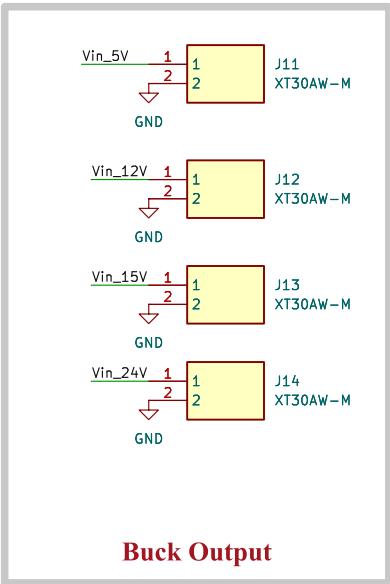
[5] Power Connectors



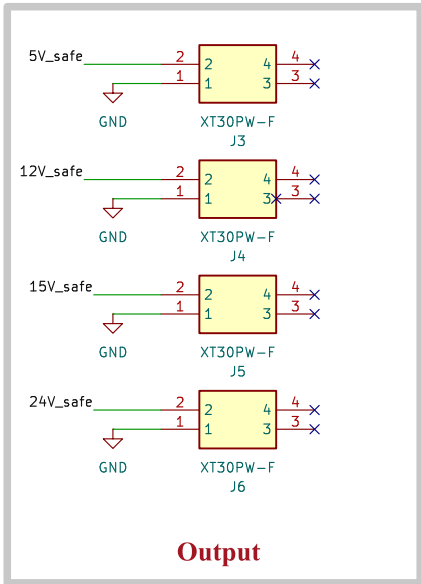
Battery connector



To Buck Modules

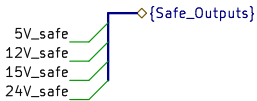
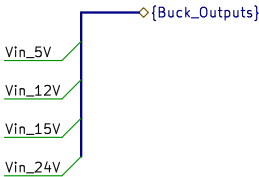


Buck Output



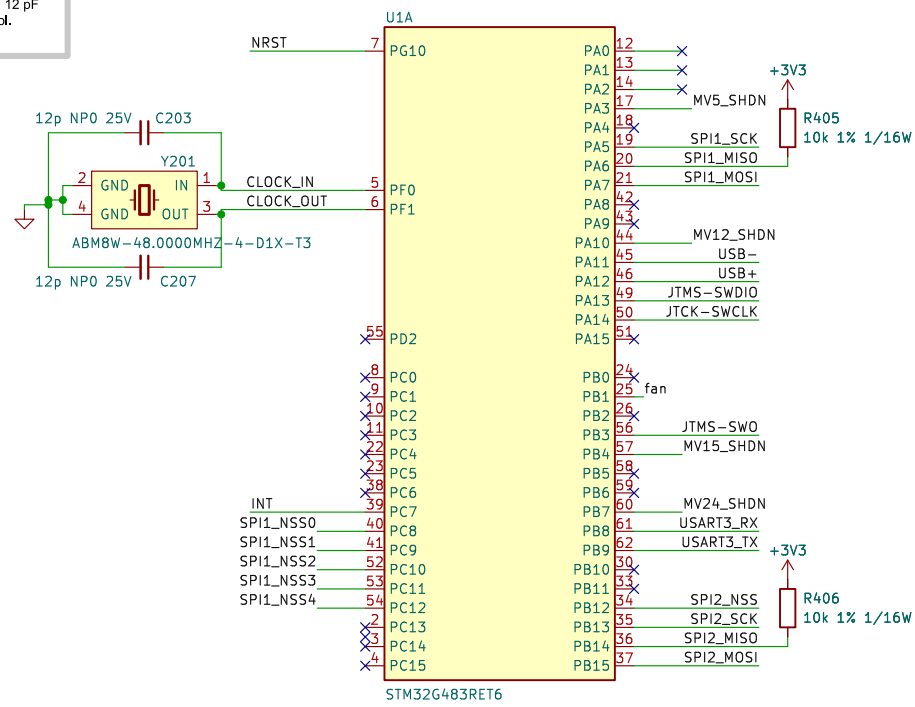
Output

DESIGN NOTE:
The battery is directly connected to Bucky via J1 and the power is directly transferred to the motors through J2. It is also connected to the bucks' input through VBuck_in.
The output of the buck are then connected to their respective voltage nets.
The same ground is used for the whole board.

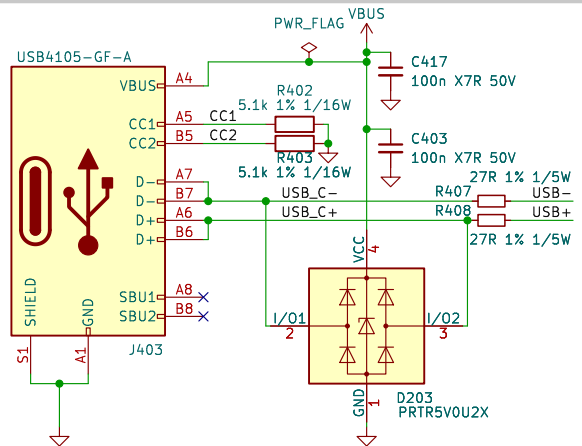


	Comments:		Company:		Variant:				
			Xplore		1.0				
			Board Name:		Project Name:				
			Bucky++		XRC				
	Sheet Title:		File Name:		Designer:		Date:		Revision:
Power Connectors		PowerConnectors.kicad_sch		Alain		Last Modified Date		B	
Sheet Path:				Reviewer:		Size:		Sheet:	
/Project Architecture/PowerConnectors/				Reviewer Name		A4		5 of 12	

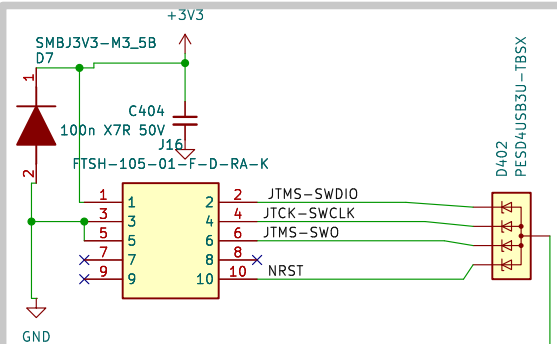
DESIGN NOTE:
Cs = 2 pF
CL = 8 pF
CL1 = CL2 = 2*(CL-CS) = 12 pF
Choose C0G/NP0 1-2% tol.



MCU I/O Domain



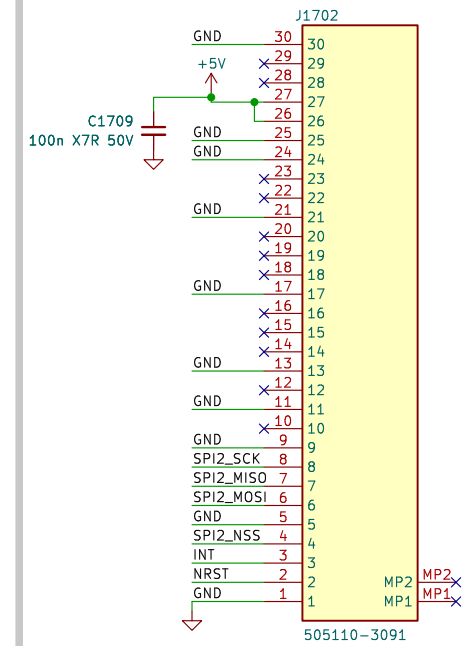
USB connector



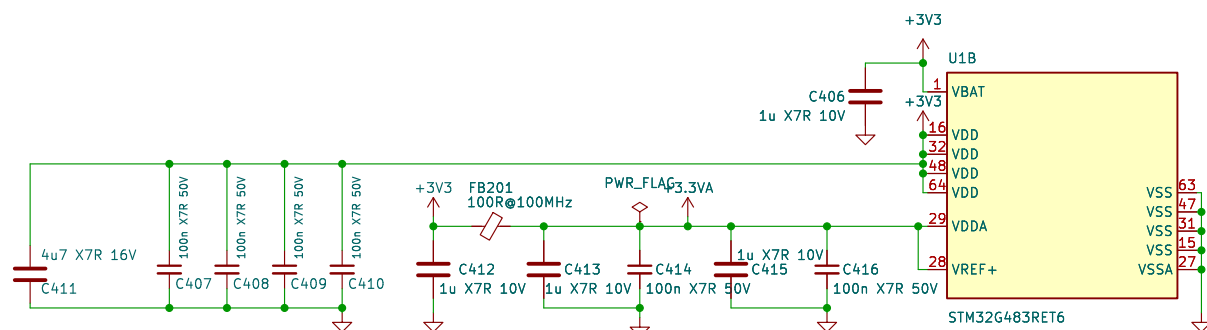
SWD Debug Connector



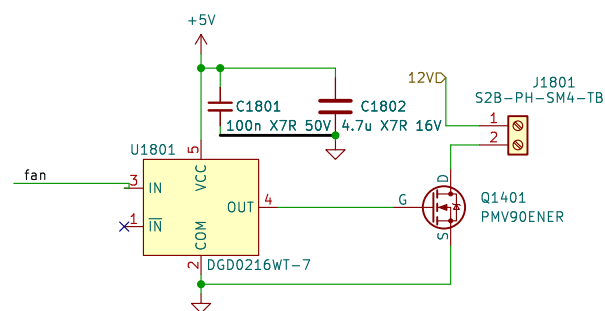
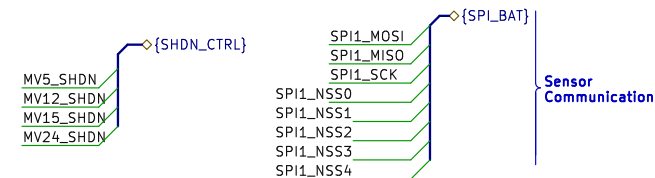
STM32 MCU that controls the power and communicate sensor data. The same model and same pins are used as GEODE to facilitate coding and developpement. Only the most important features are used here (Sensors, display, fan control, debug and basic communication connectors)



External Connector

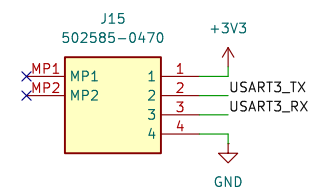


MCU Power Domain



FAN

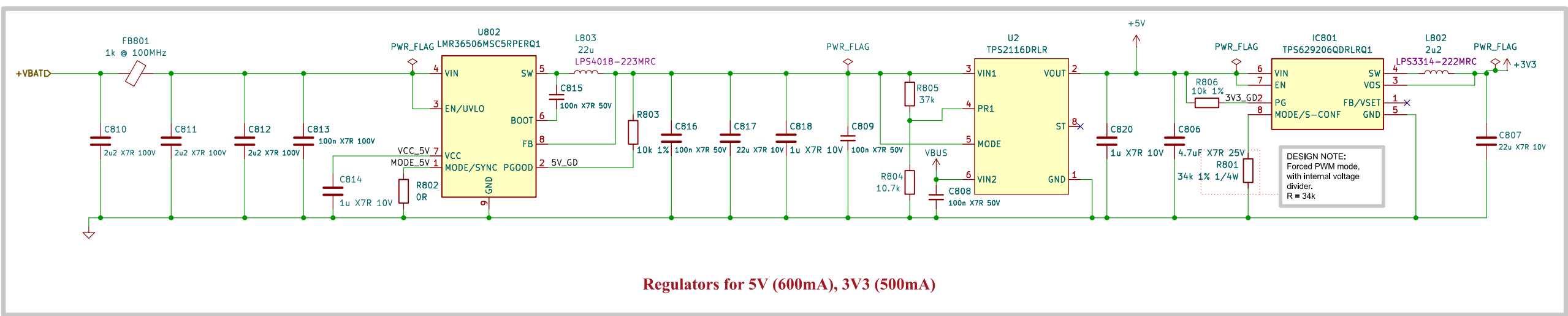
DESIGN NOTE:
The Fan is powered by the 12V output from the Bucks



USART Connector

Comments:	Company: Xplore		Variant: 1.0	
	Board Name: Bucky++		Project Name: XRC	
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
Sheet Path: /Project Architecture/MCU/		Reviewer: Reviewer Name	Size: A3	Sheet: 8 of 12

[9] Local Power Generation

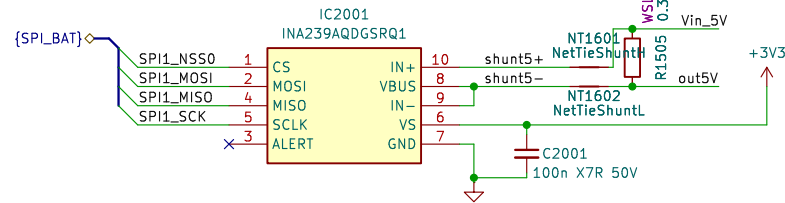


Local Power generation to power the MCU and peripherals. Schematics adapted from GEODE

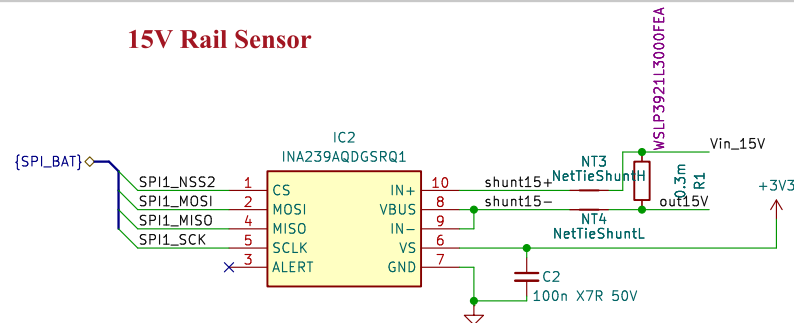
	Comments: References: Flexible I/O worked examples Flexible I/O source configuration	Company: Xplore		Variant: 1.0	
		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Local Power Generation	File Name: LocalPowerGeneration.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
Sheet Path: /Project Architecture/LocalPowerGeneration/			Reviewer: Reviewer Name	Size: A3	Sheet: 9 of 12

[10] Sensors

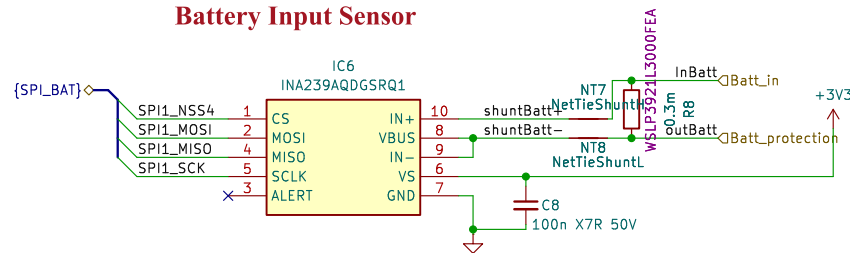
5V Rail Sensor



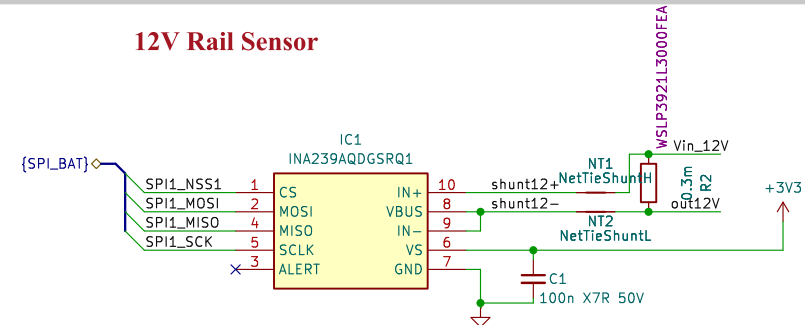
15V Rail Sensor



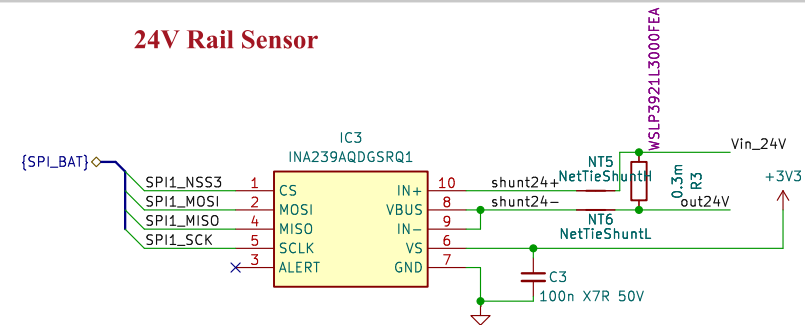
Battery Input Sensor



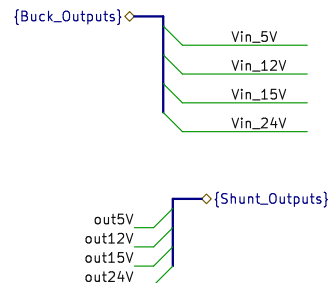
12V Rail Sensor



24V Rail Sensor



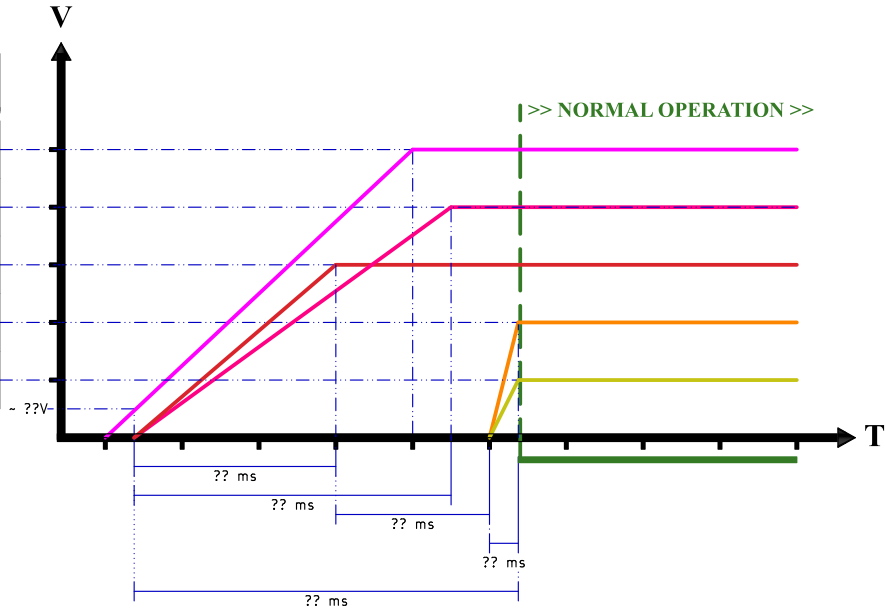
DESIGN NOTE:
One sensor for each rail is added to monitor current and voltage of the bucks' output.
One sensor is added for the input, it will help us deduce the motor power consumption. They all run on the same SPI as 8 uS is a low enough update interval. The update interval is computed by this formula:
Number of devices * bit / sensor frequency = 8 uS



Comments:	Company:		Variant:	
	Xplore		1.0	
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	Sensors		Bucky++	
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	Sensors.kicad_sch		Last Modified Date	B
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[6] Power - Sequencing

NAME	SOURCE	LEVEL
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%



	Comments:	Company: Xplore		Variant: 1.0	
		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
	Sheet Path: /Power - Sequencing/		Reviewer: Reviewer Name	Size: A4	Sheet: 6 of 12

[7] Revision History

15.11.2025 - 1.0 Revision
Variant: A

- Added schematics and e-fuse for rail outputs
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	Comments:		Company:		Variant:	
			Xplore		1.0	
			Board Name:		Project Name:	
			Bucky++		XRC	
	Sheet Title:		File Name:		Designer:	
	Revision History		Revision History.kicad_sch		Alain	
	Sheet Path:		Reviewer:		Size:	
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